

A Fourier Analysis Approach to Comparing Film Music Scores from Hans Zimmer and John Williams

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Introduction

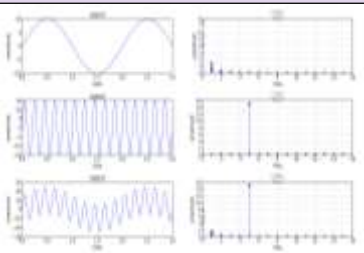


Original Sound Track, also known as OST, is the key ingredient in films to enhance the quality of the experience. They convey emotions, hints to the audience, and even work as another layer of storytelling. You can easily notice this by replacing the background soundtracks in iconic movie scenes. Imagine watching The Lord of the Rings, Star Wars, or any Marvel movie without their soundtracks. The film would be bland and often become awkward even to watch.

As a person who enjoys watching films and listening to soundtracks, my favorite composers are Hans Zimmer and John Williams. Hans Zimmer's collaboration with Christopher Nolan films is the main reason why I got into film music, and John Williams' contribution to composing such iconic film scores – Jurassic Park, Indiana Jones, Superman, Star Wars, etc. – is truly unforgettable.

There are many pieces composed by Zimmer and Williams, and sometimes I notice similarities between them. Not only from the same composer, but also from different composers, they often conveyed a likely feeling. That made me wonder, is there a mathematical reason why I love both of the composers' works? Are there any similarities between their pieces, and how can I mathematically quantify them?

Background



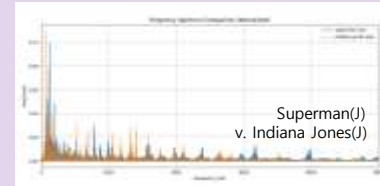
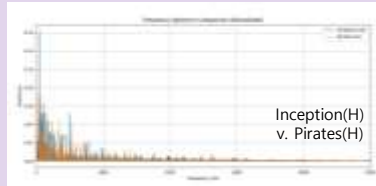
$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

- FFT (Fast Fourier Transform) can be used to convert a signal from the time domain to the frequency domain : Mathematical interpretation of sinusoidal waves into an array of magnitudes of frequencies.
- Cosine similarity can be used to calculate the similarity between the arrays by treating them as vectors.
- Using Python libraries, we can execute FFT and create graphs that show the frequency spectrum of the musical pieces and compute the cosine similarity between them.

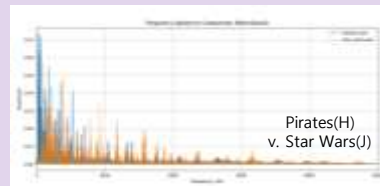
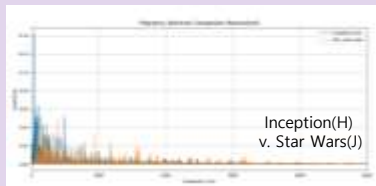
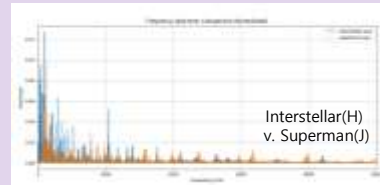
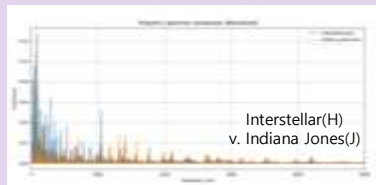
Methods

- Select three representative audio excerpts from Zimmer and Williams (6 samples total. Inception, Pirates, Interstellar by Hans Zimmer, Superman, Star wars, Indiana Jones by John Williams).
- Select two audio files for comparison.
- Conversion of audio to mono, equalizing duration, ensuring consistent sampling rates.
- Execute Fourier transform and obtain FFT magnitude vectors.
- Normalization to exclude the influence of volume.
- Plot a graph that shows the frequency spectrum of the two samples.
- Calculate the cosine similarity between the vectors.
- Repeat the process by selecting two different audio files.

Results



- Two graphs comparing two pieces composed by the same composer.
(Lt: Hans Zimmer, Rt: John Williams)



- Four graphs comparing two pieces composed by different composers.

	Inception (H)	Interstellar (H)	Pirates (H)	Star Wars(J)	Superman (J)	Indiana Jones(J)
Inception(H)	1.0	0.5299	0.5875	0.5222	0.5011	0.4626
Interstellar(H)	0.5299	1.0	0.5386	0.4831	0.5951	0.5172
Pirates(H)	0.5282	0.5386	1.0	0.4918	0.4735	0.4798
Star Wars(J)	0.5222	0.4831	0.4918	1.0	0.5516	0.5225
Superman(J)	0.5011	0.5951	0.4735	0.5516	1.0	0.5532
Indiana Jones(J)	0.4626	0.5172	0.4798	0.5225	0.5532	1.0

Discussion

- Scores coming from the same composer exhibited high cosine similarity with an average of 0.5472.
- Scores coming from different composers resulted in a cosine similarity of approximately 0.5029 on average.
- Exceptional cases between music pieces composed by different composers showed that they share similar combinations of frequencies (Interstellar v. Superman – 0.5951).
- Pitch and tone color can correspond to these results, as those elements are closely related to the mathematical concept of frequencies.
- However, other musical elements, such as rhythm or melodic contour, may be overlooked in this process since they do not have a strong relationship with frequencies.

Conclusion

- Learned that musical compositions coming from the same composer had actual similarities in frequency distributions and magnitudes.
- Mathematical tools can contemplate traditional music theories.
- This analytical technology can also be applied to genre prediction or computational musicology.
- This research mainly focused on how to approach musical comparison in a mathematical and data analytical way, so in future research, a wider range of factors can be included (bpm, rhythm, scales, harmony, melody).
- Machine learning based approaches can be made (use of AI).

References

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